

Customer Shopping Behavior Analysis

1. Project Overview :

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary :

- Rows: 3,900
- Columns: 18
- Key Features:
 - Customer demographics (Age, Gender, Location, Subscription Status)
 - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
 - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
 - Missing Data: 37 values in Review Rating column

3. EDA (Exploratory Data Analysis) using Python

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to check structure or data types and `.describe()` for summary statistics.
- Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- Feature Engineering:
 - Created `age_group` column by binning customer ages.
 - Created `purchase_frequency_days` column from purchase data.
 - Data Consistency Check: Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
 - Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

	Customer ID	Age	Gender	Item purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

Figure 1: Descriptive Statistics of the Dataset

- From above image we get statistic info like mean, min, max, frequency, 25th, 50th, 75th percentile

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                      3900 non-null   object
4   Category                            3900 non-null   object
5   Purchase Amount (USD)               3900 non-null   int64
6   Location                            3900 non-null   object
7   Size                                3900 non-null   object
8   Color                               3900 non-null   object
9   Season                              3900 non-null   object
10  Review Rating                       3863 non-null   float64
11  Subscription Status                 3900 non-null   object
12  Shipping Type                      3900 non-null   object
13  Discount Applied                   3900 non-null   object
14  Promo Code Used                    3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                     3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB
```

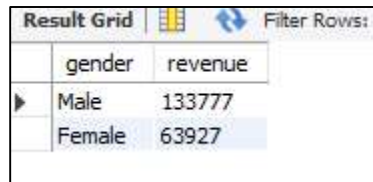
Figure 2: Dataset Information

- From above image we get info about Dtypes, columns, Null Counts

4. Data Analysis using MYSQL

We performed structured analysis in MySQL Workbench to answer key business questions:

- i. Revenue by Gender – Compared total revenue generated by male vs. female customers.



The screenshot shows the MySQL Workbench interface with a 'Result Grid' tab. It displays a table with two columns: 'gender' and 'revenue'. The data is as follows:

gender	revenue
Male	133777
Female	63927

- ii. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
22	62
24	88
29	94
32	79
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- iii. Top 5 Products by Rating – Found products with the highest average review ratings.

item_purchased	Average_product_Rating
Gloves	3.9
Sandals	3.84
Hat	3.8
Boots	3.8
Jacket	3.79

- iv. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

shipping_type	round(avg(purchase_amount), 2)
Express	60.56
Standard	58.12

- v. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

subscription_status	total_customers	avg_spend	total_revenue
Yes	896	59.27	53105
No	2425	59.63	144599

- vi. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases

item_purchased	discount_rate
Hat	50.74
Sweater	48.87
Sneakers	48.84
Boots	48.33
Coat	47.86
Pants	47.65
Jeans	46.09
Dress	45.83
Shorts	44.53
Jewelry	44.20

- vii. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history

customer_segment	Number of Customers
loyal	2651
returning	599
new	71

- viii. Top 3 Products per Category – Listed the most purchased products within each category.

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	138
2	Accessories	Hat	136
3	Accessories	Sunglasses	136
1	Clothing	Pants	149
2	Clothing	Shirt	145
3	Clothing	Dress	144
1	Footwear	Sandals	138
2	Footwear	Sneakers	129
3	Footwear	Shoes	128
1	Outerwear	Jacket	143
2	Outerwear	Coat	140

- ix. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	repeat_buyers
Yes	816
No	2149

- x. Revenue by Age Group – Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	51686
Middle Age	50180
Adult	48402
Senior	47436

5. Dashboard in Power BI



Figure : Power BI Dashboard